

Project #12: Adaptive XR Neurofeedback/Biofeedback Platform for Rehabilitation and Exercise Training

1. Project Description

This graduation project aims to develop an adaptive extended reality (XR) platform that combines virtual reality (VR), mixed reality (MR), biofeedback, and neurofeedback for rehabilitation and exercise training. The proposed system uses an XR headset, electroencephalography (EEG), electromyography (EMG), inertial measurement units (IMUs), and motion capture cameras to create an immersive and measurable training environment. During exercise, the platform continuously monitors movement quality, muscle activation, attention, concentration, and workload, then adapts the virtual or mixed-reality task in real time to improve engagement, motor learning, and performance.

The project addresses the growing need for intelligent rehabilitation and training systems that go beyond simple exercise display. Instead of delivering fixed tasks, the proposed platform forms a closed-loop human-centered mechatronic system. It collects multimodal physiological and kinematic signals, processes them in real time, estimates the user state, and feeds the results back through visual cues, performance scores, adaptive difficulty, and neurofeedback-guided exercise progression. Such a platform is valuable for upper-limb rehabilitation, gait and balance training, sports exercise assessment, and future personalized digital therapy applications.

2. Main Objectives

- Develop immersive VR and MR exercise scenarios using a modern XR platform and interactive software environment.
- Integrate EEG, EMG, IMUs, and motion-capture cameras into one synchronized measurement system.
- Estimate user concentration, focus, muscle engagement, movement quality, fatigue, and task performance in real time.
- Provide biofeedback and neurofeedback through avatars, graphs, indicators, and adaptive exercise cues.
- Adapt exercise difficulty, repetition level, and task progression to enhance neuroplasticity and training outcomes.
- Store session data and generate objective reports for clinical follow-up, sports assessment, and research use.



Figure 1. Conceptual view of the adaptive XR rehabilitation and training platform with EEG, EMG, IMUs, motion capture, and real-time analysis software.

3. Proposed System Architecture

The overall architecture begins with the user performing a rehabilitation or exercise task in an XR session. The sensing layer measures brain activity using EEG, muscle activation using EMG, movement kinematics using IMUs and optical motion capture, and optional force or grip interaction when needed. All acquired data are synchronized and streamed to a processing unit where filtering, feature extraction, and AI-based assessment algorithms are executed. The system estimates the user state, including attention, focus, muscle activation quality, range of motion, fatigue level, and movement correctness. Based on these estimates, the XR environment modifies the task and presents feedback in a personalized way.

System Architecture of the Adaptive XR Neurofeedback Rehabilitation Platform

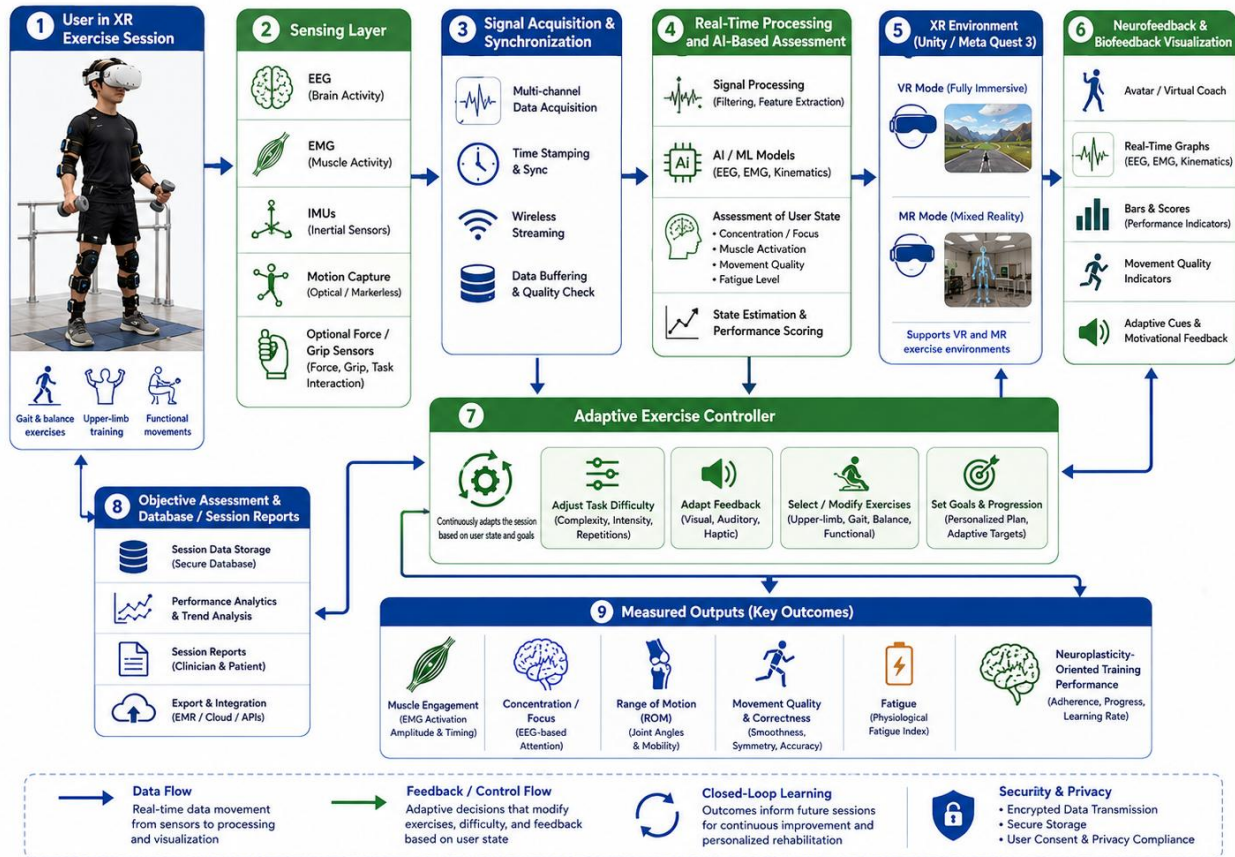


Figure 2. System architecture showing sensing, synchronization, AI-based assessment, adaptive exercise control, neurofeedback visualization, and outcome reporting.

4. Technical Workflow

Step 1 – Calibration and Setup: The user wears the XR headset and sensors, and the system calibrates the EEG, EMG, IMUs, and motion-capture modules.

Step 2 – Exercise Execution: The user performs upper-limb, gait, balance, or functional exercises in VR or MR while all signals are recorded.

Step 3 – Real-Time Processing: Signal-processing and machine-learning modules extract features and estimate concentration, muscle engagement, workload, and movement quality.

Step 4 – Adaptive Feedback: The platform provides scores, graphs, virtual coaching cues, and corrective feedback while adjusting exercise intensity, repetitions, or task complexity.

Step 5 – Objective Assessment: At the end of the session, the system stores the data and generates performance trends, personalized reports, and progression indicators.

The closed-loop nature of the system is one of its key strengths. Rather than using static tasks, the platform uses neurofeedback and biofeedback to continuously reshape the session according to the user's behavior. This approach supports more engaging exercise delivery, improved objective assessment, better personalization, and stronger potential for enhancing neuroplasticity and motor recovery.

5. Adaptive Neurofeedback and Measured Outcomes

In the proposed platform, neurofeedback is used to reflect the user's mental state and engagement, while biofeedback reflects physical effort and movement execution. This integration enables a richer understanding of how the brain, muscles, and movement interact during exercise. The training system can therefore encourage better concentration, balanced muscle recruitment, improved exercise form, and gradual progression toward individualized goals.

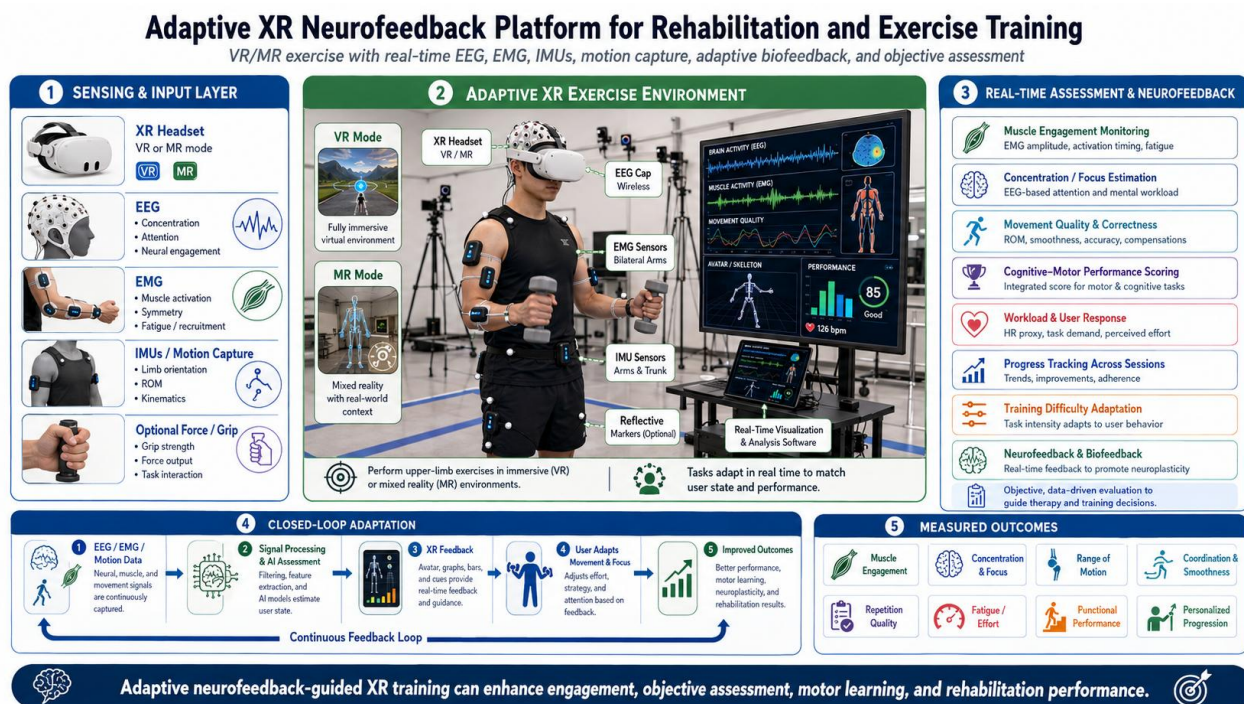


Figure 3. Integrated infographic summarizing sensing, adaptive XR exercise delivery, real-time assessment, neurofeedback, closed-loop adaptation, and key measured outcomes.

6. Student Tasks and Expected Deliverables

- Develop and refine VR/MR exercise scenes and user interfaces for rehabilitation and training tasks.
- Interface EEG, EMG, IMUs, and motion-capture systems and build a synchronized acquisition pipeline.
- Implement real-time dashboards for concentration, muscle activity, movement quality, repetition count, and performance scoring.
- Design adaptive exercise logic that modifies difficulty, feedback mode, and progression according to user state.
- Evaluate the system using healthy participants and report metrics such as range of motion, symmetry, smoothness, repetition quality, attention level, and fatigue.
- Prepare a complete technical report, presentation, and demonstration of the integrated platform.

Previous related project video: <https://www.youtube.com/watch?v=huUIYgq6ImQ>

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